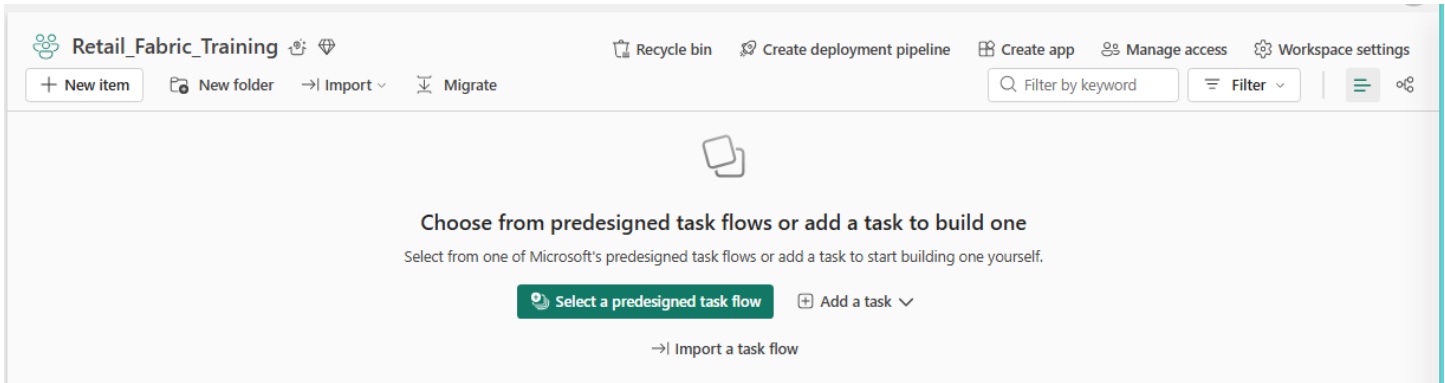
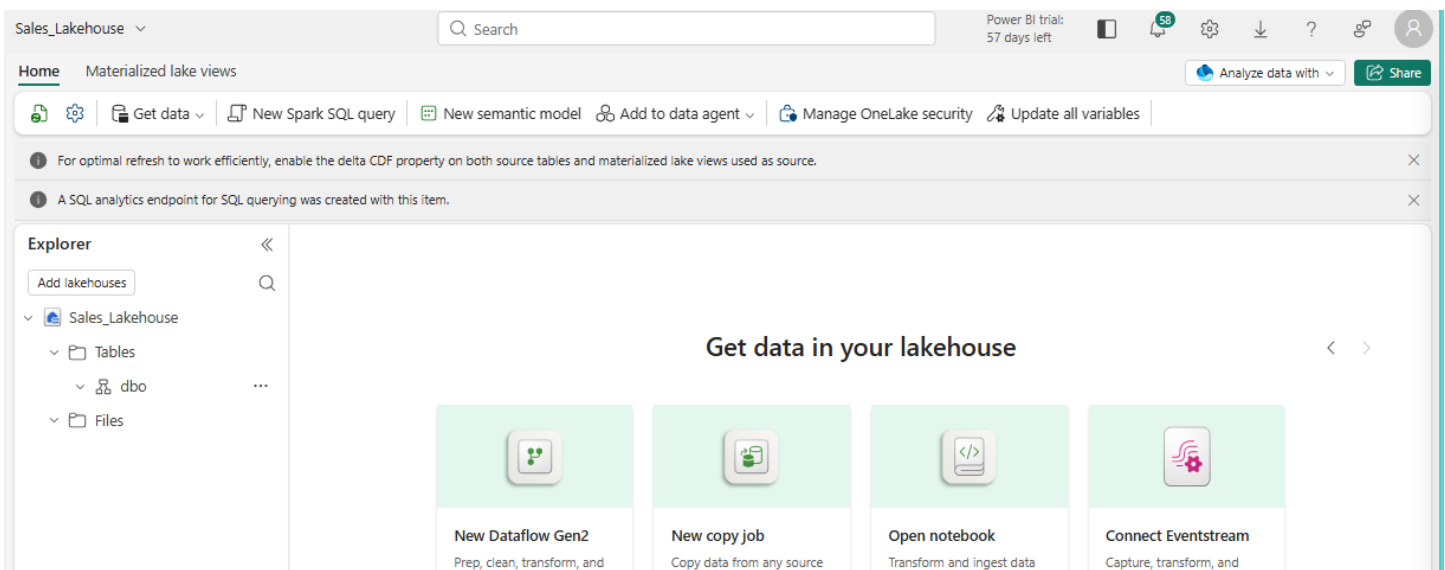


1. Create Fabric Workspace



2. Create Lakehouse



Q1. What is a Lakehouse?

A Lakehouse combines the flexibility of a data lake with the structured data management and analytics capabilities of a data warehouse.

Q2. Purpose of Files

Files stores raw data such as CSV, JSON, Excel, etc.

Q3. Purpose of Tables

Tables contain structured data that can be queried and analyzed.

Q4. What is SQL analytics endpoint?

It allows you to query Lakehouse tables using SQL.

3. Copy CSV to Lakehouse

Sales_Lakehouse

Search

Power BI trial: 56 days left

58

Settings

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HomeMaterialized lake views

Get data

New Spark SQL query

New semantic model

Add to data agent

Manage OneLake security

Update all variables

For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views used as source.

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

Add lakehouses

Sales_Lakehouse

Tables

dbo

Files

Fabric_Sales_50...

Retail_Fabric_Training > Sales_Lakehouse > Files > Fabric_Sales_500.xlsx (preview)

View settings

	() _c0	() _c1	ABC _c2	ABC _c3	ABC _c4	ABC _c5	() _c6	() _c7
1	OrderID	OrderDate	CustomerID	Product	Category	Region	Quantity	SalesAmo
2	1001	5/21/26	C016	Notebook	Office Supplies	South	2	6.6
3	1002	2/14/26	C038	Keyboard	Electronics	West	2	148.28
4	1003	4/30/26	C033	Laptop	Electronics	North	1	720.27
5	1004	8/3/26	C015	Bookshelf	Furniture	West	4	972.8
6	1005	1/4/26	C049	Filing Cabinet	Furniture	South	5	1088
7	1006	3/21/26	C014	Pen Set	Office Supplies	East	7	82.67
8	1007	7/3/26	C023	Monitor	Electronics	East	2	603.16
9	1008	3/5/26	C025	Stapler	Office Supplies	North	1	15.95

Explorer

Add lakehouses

Sales_Lakehouse

Sales_Lakehouse - Retail_Fabric_Training

dbo

Fabric_Sales_500

Files

Fabric_Sales_500

Table view

Showing 501 rows

	ABC Prop_0	ABC Prop_1	ABC Prop_2	ABC Prop_3	ABC Prop_4	ABC Prop_5	ABC Prop_6	ABC Prop_7
1	OrderID	OrderDate	CustomerID	Product	Category	Region	Quantity	SalesAmount
2	1001	2026-05-21	C016	Notebook	Office Supplies	South	2	6.6
3	1002	2026-02-14	C038	Keyboard	Electronics	West	2	148.28
4	1003	2026-04-30	C033	Laptop	Electronics	North	1	720.27
5	1004	2026-08-03	C015	Bookshelf	Furniture	West	4	972.8
6	1005	2026-01-04	C049	Filing Cabinet	Furniture	South	5	1088
7	1006	2026-03-21	C014	Pen Set	Office Supplies	East	7	82.67
8	1007	2026-07-03	C023	Monitor	Electronics	East	2	603.16
9	1008	2026-03-05	C025	Stapler	Office Supplies	North	1	15.95
10	1009	2026-07-05	C037	Bookshelf	Furniture	South	5	1329.4
11	1010	2026-04-27	C050	Pen Set	Office Supplies	East	2	11.84
12	1011	2026-07-14	C018	Monitor	Electronics	West	4	1829.08
13	1012	2026-07-01	C014	Notebook	Office Supplies	East	6	29.7
14	1013	2026-03-29	C035	Pen Set	Office Supplies	South	2	34.36
15	1014	2026-04-23	C044	Keyboard	Electronics	East	8	402.88

Succeeded (1 sec 312 ms)

Columns 8 Rows 501

4. Validate the Lakehouse Sales table

The screenshot shows a SQL query editor interface. On the left is the 'Explorer' pane with a tree view containing 'Warehouses', 'Sales_Lakehouse' (with sub-items 'Schemas' and 'Security'), 'Queries' (with sub-item 'My queries'), and 'SQL query 1' (selected). The main editor area has a tab for 'SQL query 1' and a toolbar with 'Run', 'Save as', 'Save as view', 'Copy query', and 'Create rule'. The query text is: `1 SELECT COUNT(*) AS RecordCount FROM dbo.Sales;`. Below the query, the 'Results' tab is active, displaying a table with one row:

	RecordCount
1	500

.

The screenshot shows the same SQL query editor interface. The query text is: `1 SELECT COLUMN_NAME, DATA_TYPE FROM INFORMATION_SCHEMA.COLUMNS
2 WHERE TABLE_NAME = 'Sales'
3 ORDER BY ORDINAL_POSITION;`. The 'Results' tab is active, displaying a table with two columns: 'COLUMN_NAME' and 'DATA_TYPE'. The data is as follows:

	COLUMN_NAME	DATA_TYPE
1	OrderID	varchar
2	OrderDate	varchar
3	CustomerID	varchar
4	Product	varchar
5	Category	varchar
6	Region	varchar
7	Quantity	varchar
8	SalesAmount	varchar

.

Explorer

Warehouses

Sales_Lakehouse

Schemas

Security

Queries

My queries

SQL query 1

Shared queries

SQL query 1

Run Save as Save as view Copy query Create rule Explain query Fix query errors

```
1 SELECT
2     COUNT(*) AS TotalRows,
3     COUNT(OrderID) AS OrderID_Count,
4     COUNT(OrderDate) AS OrderDate_Count,
5     COUNT(CustomerID) AS CustomerID_Count,
6     COUNT(Product) AS Product_Count,
7     COUNT(Category) AS Category_Count,
8     COUNT(Region) AS Region_Count,
9     COUNT(Quantity) AS Quantity_Count,
10    COUNT(SalesAmount) AS SalesAmount_Count
11 FROM dbo.Sales;
12
13
```

Messages Results

Search

	TotalRows	OrderID_Count	OrderDate_Count	CustomerID_Count	Product_Count	Category_Count	Region_Count	Quantity_Count	SalesAmount_Count
1	500	500	500	500	500	500	500	500	500

Explorer

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Shared queries

SQL query 1

Run Save as Save as view Copy query Create rule Explain query Fix query errors

```
1 SELECT
2     OrderID,
3     COUNT(*) AS DuplicateCount
4 FROM dbo.Sales
5 GROUP BY OrderID
6 HAVING COUNT(*) > 1;
```

Messages Results

Search

ABC OrderID	DuplicateCount
-------------	----------------

Explorer

Warehouses

Sales_Lakehouse

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SQL query 1

Shared queries

SQL query 1

Run Save as Save as view Copy query Create rule Explain query Fix query errors

```
1 SELECT TOP 10 *
2 FROM dbo.Sales;
```

Messages Results

Search

	ABC OrderID	ABC OrderDate	ABC CustomerID	ABC Product	ABC Category	ABC Region	ABC Quantity	ABC SalesAmount
1	1001	2026-05-21	C016	Notebook	Office Supplies	South	2	6.6
2	1002	2026-02-14	C038	Keyboard	Electronics	West	2	148.28
3	1003	2026-04-30	C033	Laptop	Electronics	North	1	720.27
4	1004	2026-08-03	C015	Bookshelf	Furniture	West	4	972.8
5	1005	2026-01-04	C049	Filing Cabinet	Furniture	South	5	1088
6	1006	2026-03-21	C014	Pen Set	Office Supplies	East	7	82.67
7	1007	2026-07-03	C023	Monitor	Electronics	East	2	603.16
8	1008	2026-03-05	C025	Stapler	Office Supplies	North	1	15.95
9	1009	2026-07-05	C037	Bookshelf	Furniture	South	5	1329.4

AutoSave: On (Saved) Succeeded (1 sec 523 ms) Copilot completions: Off Code assistance: objects refreshed (refreshed 4m ago) Columns: 8 Rows: 10

Q1. How many records are present?

There are 500 records in the Sales table.

Q2. What is the data type of OrderID?

The data type of OrderID is VARCHAR.

Q3. What is the data type of Quantity?

The data type of Quantity is VARCHAR.

Q4. What is the data type of SalesAmount?

The data type of SalesAmount is VARCHAR.

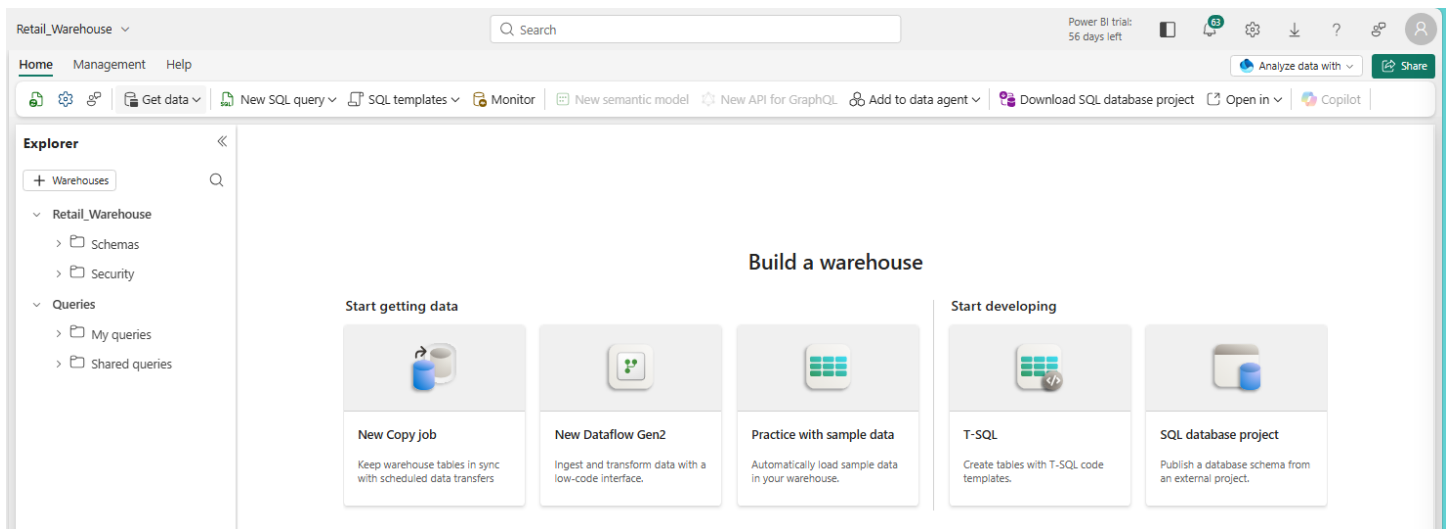
Q5. Are there any null values?

No, there are no null values in the Sales table.

Q6. Are there duplicate OrderID values?

No, there are no duplicate OrderID values.

5. Create the Warehouse



Retail_Fabric_Training

Recycle bin
 Create deployment pipeline

+ New item

New folder

Import

Migrate

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow
 Add a task

→ Import a task flow

	Name	Status	Type	Task	Owner	Refreshed	Next refresh
	CopyJob_1		Copy job	—	Hemanth V	—	—
	CopyJob_2		Copy job	—	Hemanth V	—	—
	Retail_Warehouse		Warehouse	—	Hemanth V	—	—
	Sales_Lakehouse		Lakehouse	—	Hemanth V	—	—
	Sales_Lakehouse		SQL analytics ...	—	Hemanth V	—	—

Q1. What is a Warehouse?

A Warehouse is a structured data storage system designed for storing, managing, and analyzing large amounts of data using SQL.

Q2. How is a Warehouse different from a Lakehouse?

A Warehouse is mainly designed for structured data and SQL analytics, while a Lakehouse can handle both structured and unstructured data.

Q3. What type of workloads are commonly performed in a Warehouse?

Warehouses are commonly used for SQL queries, reporting, dashboards, business intelligence, and data analysis.

6. Copy Sales from Lakehouse to Warehouse

The screenshot shows the Azure Data Studio interface with a data copy job in progress. The job is named 'CopyJob_3'. The source is 'Sales_Lakehouse Lakehouse' and the destination is 'Retail_Warehouse Warehouse'. The job status is 'Succeeded'. The results table shows the following data:

Source → Destination	Status	Rows read	Rows written	Duration	Run start	Run end
dbo.Sales → dbo.Sales	Succeeded	500	500	42 sec	10/1/2026, 11:19:23 PM	10/1/2026, 11:20:06 PM

7. Run SQL queries in the warehouse

The screenshot shows the Azure Data Studio interface with a SQL query being run in the 'Retail_Warehouse' warehouse. The query is named 'SQL query 1'. The query text is:

```
1 SELECT COUNT(*) AS RecordCount
2 FROM dbo.Sales;
3
```

The results table shows the following data:

RecordCount
500

Retail_Warehouse

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 SQL templates

 Monitor

 New semantic model

 New API for

Explorer

Warehouses

Retail_Warehouse

Schemas

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Queries

My queries

SQL query 1

Shared queries

SQL query 1

RunSave asSave as viewCopy queryCreate ruleExplain query

```
1 SELECT
2     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales
3 FROM dbo.Sales;
4 
```

MessagesResults

e% TotalSales

1406260.56

Retail_Warehouse

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SQL query 1

Shared queries

SQL query 1

RunSave asSave as viewCopy queryCreate rule

```
1 SELECT
2     SUM(CAST(Quantity AS INT)) AS TotalQuantity
3 FROM dbo.Sales;
4 
```

MessagesResults

123 TotalQuantity

12326

Retail_Warehouse

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SQL query 1

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SQL query 1

RunSave asSave as viewCopy queryCreate ruleExplain queryFix query errors

```
1 SELECT
2     Region,
3     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales
4 FROM dbo.Sales
5 GROUP BY Region
6 ORDER BY TotalSales DESC;
7 
```

MessagesResults

	ABC Region	e* TotalSales
1	North	129494.97
2	South	108117.73
3	West	89733.63
4	East	78914.23

Retail_Warehouse

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SQL query 1

RunSave asSave as viewCopy queryCreate ruleExplain queryFix query errors

```
1 SELECT
2     Category,
3     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales
4 FROM dbo.Sales
5 GROUP BY Category
6 ORDER BY TotalSales DESC;
7 
```

MessagesResults

	ABC Category	e* TotalSales
1	Electronics	247387.98
2	Furniture	146718.24
3	Office Supplies	12154.34

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SQL query 1

RunSave asSave as viewCopy queryCreate ruleExplain queryFix query errors

```
1 SELECT TOP 5
2     Product,
3     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales
4 FROM dbo.Sales
5 GROUP BY Product
6 ORDER BY TotalSales DESC;
7
```

MessagesResults

	abc Product	e* TotalSales
1	Laptop	141280.51
2	Desk	46358.50
3	Monitor	45697.46
4	Office Chair	39158.18
5	Filing Cabinet	31178.03

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SQL query 1

RunSave asSave as viewCopy queryCreate ruleExplain queryFix query errors

```
1 SELECT
2     CustomerID,
3     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales
4 FROM dbo.Sales
5 GROUP BY CustomerID
6 ORDER BY TotalSales DESC;
7
```

MessagesResults

	abc CustomerID	e* TotalSales
1	C006	17156.53
2	C007	15001.40
3	C005	14862.95
4	C010	14848.78
5	C028	14494.26
6	C022	14433.71
7	C033	13719.18
8	C019	13613.78
9	C024	13446.16

AutoSave: On (Saved)

Succeeded (1 sec 559 ms)

Copilot completions: Off

Code assistance: objects refreshed (refreshed 10m ago)

8. Compare Lakehouse and Warehouse

The screenshot shows the Azure Data Studio interface with a SQL query titled "SQL query 1". The query compares data from a Lakehouse and a Warehouse. The query is as follows:

```
1 SELECT
2     'Lakehouse' AS Source,
3     COUNT(*) AS RecordCount,
4     SUM(CAST(Quantity AS INT)) AS TotalQuantity,
5     SUM(CAST(SalesAmount AS DECIMAL(18,2))) AS TotalSales,
6     COUNT(DISTINCT CustomerID) AS CustomerCount,
7     COUNT(DISTINCT Product) AS ProductCount
8 FROM [Sales_Lakehouse].[dbo].[Sales]
9
10 UNION ALL
11
12 SELECT
13     'Warehouse' AS Source,
14     COUNT(*) AS RecordCount,
```

The results are displayed in a table with the following columns: Source, RecordCount, TotalQuantity, TotalSales, CustomerCount, and ProductCount. The results show that the Lakehouse and Warehouse have identical data for all metrics.

	Source	RecordCount	TotalQuantity	TotalSales	CustomerCount	ProductCount
1	Lakehouse	500	2326	406260.56	50	15
2	Warehouse	500	2326	406260.56	50	15

Q1. Are the record counts equal?

Yes, the record counts are equal in both the Lakehouse and Warehouse.

Q2. Is the total sales amount equal?

Yes, the total sales amount is the same in both.

Q3. Is the total quantity equal?

Yes, the total quantity is the same in both.

Q4. If the results are different, what could be the reason?

Differences could happen because of missing records, incorrect data types, duplicate records, or errors during data transfer.

Q5. How would you troubleshoot the difference?

I would compare the record counts, check for duplicates and NULL values, verify the data types, and check the copy activity for any errors.